

PROGRAM 5

CSC 1300, SPRING 2026

SUPERHERO CONDO



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Due Date

Due Date: Monday, April 27, 2026, You may submit late up to 2 days with 2 points off per day late.

What to Turn In:

You must zip all the following files in a zip file and then upload the zip file to ilearn. **-5 points for not including all the files and submitting them in a zipped file!**

- driver.cpp
- prog5.h
- functions.cpp
- hero.txt – this should be a text file with at least FIVE super heroes that **you created with your program**
- prog5report

Program Report

Fill out this program report and take a screenshot of the confirmation page after submitting and name it **prog5report**. This is similar to the lab reports but for programming assignments instead.

Program Report Link: https://tntech.co1.qualtrics.com/jfe/form/SV_5vhCYdoB1bhJXM2

What You Can't Use in This Program

If you do any of the following, you will receive a 0 on some or all of the assignment and possibly get charged with an academic integrity violation (this is very serious).

1. You may not use code or syntax in your assignment **that has not been taught** in CSC 1300 class **without referencing your source**. How do you reference your source? By providing a link/URL in a multi-line comment in the section of code where you use “non-1300” code. Otherwise, this will be considered plagiarism. It is safer to use course materials (slides & example programs) to help you create solutions to the assignments.
2. **Do not use the following programming constructs for Program 5**. If you create a class or classes or use any STL classes such as map, vector, algorithm, array, etc., you will get a zero on the assignment. We need to make sure you can do the basics first before adding these types of constructs.

3. **You may not (ever) use the following** in your programming & lab assignments in CSC 1300:
 - while(true) or while(1)
 - goto statements
 - recursion
 - exit()
 - ternary operators
 - Any syntax requiring a '?'
 - lambda functions
 - try-catch
4. You **may NOT use generative AI (genAI)** in any way for this programming assignment including ChatGPT, Copilot, etc. Not sure if something is GenAI? Ask a TA or instructor!
5. **This is not a group effort.** Programming assignments are like exams in this class and should be **your own work**. You may discuss the assignment with other students in the class but you are not allowed to share/send your code to any other person (friend, past/present students, parent, etc.) or view/receive code from anyone (unless they are a TA or instructor for this class). This is considered **cheating** in this class and you will get a **zero** on the assignment (even if you are the one who wrote the code and sent it to someone else). Get help from TAs, tutors & instructors when you get stuck!

Description

Your assignment is to write a program for a superhero condo landlord. Your program should ask the landlord how many superheroes the condo can hold and dynamically allocate an array of **SuperHero** structures. Information about each hero needs will be kept and manipulated in this program. The user should be able to load hero information from any file he or she chooses, add heroes manually, delete a hero, print hero information to the screen, or print rent information.

PROGRAM SPECIFICATIONS

PROG5.H HEADER FILE

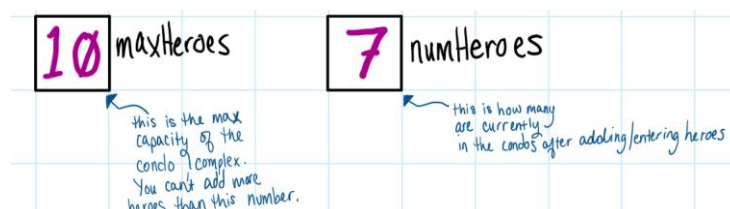
The header file should contain all #includes, namespace, structure declaration (see below), and function prototypes. Also, don't forget the include guards.

Create a structure called **Heroes** that has following members:

- The name of the superhero
- The description of the superhero
- If the hero is dangerous (**bool** variable)
- The cost of this hero's rent per month
- The cost of damage that this hero typically has per month
- The number of years (allow partial years) that the hero has lived in the condo

DRIVER.CPP (CONTAINS THE MAIN FUNCTION)

Ask the landlord how many superheroes can live in his/her condo complex. This is the maximum number of heroes you can add to your program. Also create a variable that will keep up with the current number of heroes living in the complex (the number that has been added during runtime of the program).



Dynamically allocate an array of **Heroes** of the maximum capacity size.

heroName	danger	heroName	danger	heroName	danger	heroName	danger	heroName	danger	heroName	danger	heroName	danger	heroName	danger	heroName	danger
Superman	0	Wonder Woman	1	Cyborg	0	Elongated Man	0	Zatanna	1	Black Canary	1	April Cockett	1				
Superman is...	costRent	Wonder Woman is...	costRent	Large portions...	costRent	As a teenager...	costRent	Zatanna, mmm...	costRent	A world class...	costRent	April is a...	costRent				
costDamage	rentHours	costDamage	rentHours	costDamage	rentHours	costDamage	rentHours	costDamage	rentHours	costDamage	rentHours	costDamage	rentHours				
200.00	5.00	300.00	.50	150.00	4.00	0.00	10.00	98.00	1.50	44.00	6.00	0.00	4.00				

The main function will display a menu of five options:

```
What would you like to do?
1. Enter some superheroes
2. Delete a superhero
3. Print all superheroes
4. Print rent details.
5. End Program
Enter 1, 2, 3, 4, or 5.
CHOICE:
```

Make sure you always validate all user choices in the whole program before proceeding to do what the user wants. Each menu choice will call a function that you will create.

If the user chooses option 1, then your program will call the **enterHeroes ()** function. If the user chooses option 2, then your program will call the **deleteHero ()** function. If the user chooses option 3, then your program will call the **printHeroes ()** function. If the user chooses option 4, then your program will call the **printRentDetails ()** function. If the user chooses option 5, then your program will ask the user if they wish to save their hero tenants to a file.

Would you like to save your superheroes list to a file? (y or n)

If they choose 'y', then your program should call the **saveToFile ()** function and then end. If they choose 'n', then your program should just end. (Make sure to validate that they entered either 'y' or 'n').

Don't forget to release any memory that was dynamically allocated at the end of the main function!

FUNCTIONS.CPP

You are given a source file named **functions_given.cpp** that contains part of the **enterHeroes()** function.

enterHeroes() Function

The **enterHeroes ()** function takes three parameters:

- The maximum number of heroes that the condo can have
- The current number of heroes currently living in the code (stored in the **Heroes** array)
- The **Heroes** array

The function will return the new (updated) number of heroes in the array after adding. Before trying to add any heroes, this function should first check to make sure the number of heroes isn't already at the maximum number. Because if it is, then your program should not add any heroes, but should instead tell the user that the condo is at full capacity and that they are not able to add any more heroes. Then the function should end.

Example:

```
Sorry! Your complex can only hold 7 heroes.
You cannot add more.
```

Otherwise, your program will display a menu asking the user if they would like to do one of the following:

```
What do you want to do?
    1. Load the heroes from a file.
    2. Enter a hero manually.
    Choose 1 or 2.
CHOICE:
```

Validate the user's choice.

READ FROM FILE

THIS PART IS WRITTEN FOR YOU!! If the user chooses **option 1**, then the program will ask the user what the name of the file is that they would like to load the heroes from and then it will read in the data from the text file and fill up the hero array.

READ FROM USER MANUALLY

If the user chooses **option 2**, then you will want to start a loop so that the user can add multiple heroes manually without returning to the main menu. Ask the user for the hero data and make sure you place each bit of information in the correct place in the **Heroes** array.

Example of option 2 is below with user input highlighted in yellow.

SUPERHERO NAME: Robin

DESCRIPTION: Richard John "Dick" Grayson was once Batman's first sidekick Robin, before going on to become Nightwing. The youngest in a family of acrobats known as the "Flying Graysons," Dick watched as a mafia boss killed his parents in order to extort money from the circus that employed them (whose name was later revealed to be Tony Zucco). Bruce Wayne, secretly the vigilante Batman, took him in as his legal ward after witnessing their deaths, and eventually as his sidekick, Robin.

IS IT A DANGEROUS HERO? (y or n): y

How much does Robin pay for rent per month?

RENT PRICE: \$987.57

What is the typical cost of damage Robin has each month?

DAMAGE COST: \$99.34

How many years has Robin lived in your condo?

YEARS: 2.75

The Robin has been added.

Want to add more heroes? (y or n) y

SUPERHERO NAME:

Then, increment the number of current heroes by one. Then, ask the user if they want to add another hero (if there is room in the array). If they do, then repeat this option. If not, then just return the number of heroes.

deleteHero() Function

The `deleteHero()` function has two parameters: the current number of heroes in the `Heroes` array and the `Heroes` array. This function returns the new (updated/reduced) number of heroes.

First, this function will say “**The following is a list of all the heroes living in your condo complex:** ” and then it will say the name of each superhero. Then, your program will ask the user “**Which hero are you kicking out of your complex?**”

Example below of user deleting Robin with user input highlighted in yellow:

```
What would you like to do?
1. Enter some superheroes
2. Delete a superhero
3. Print all superheroes
4. Print rent details.
5. End Program
Enter 1, 2, 3, 4, or 5.
```

CHOICE: 2

The following is a list of all the heroes living in your condo complex:

```
Superman
Wonder Woman
Cyborg
Elongated Man
Zatanna Zatara
Black Canary
Robin
April Crockett
```

Which hero are you kicking out of your complex?

SUPERHERO NAME: Robin

Your program should then read in the name and place it in a variable. Then, this function will call the `moveArrayElements` function, which will take care of removing the hero. The `moveArrayElements` function returns a Boolean value to tell if the hero was removed or not. Check to see if the hero was removed. If it was, then decrement the number of current heroes and print out “You have removed [insert hero name here].” Then return the new number of heroes.

Which hero are you kicking out of your complex?

SUPERHERO NAME: Robin

You have removed Robin.

If your program could not find the name, then you should print out “Sorry, a hero by the name [enter name here] could not be found”.

The following is a list of all the heroes living in your condo complex:

```
Superman
Wonder Woman
Cyborg
Elongated Man
Zatanna Zatara
Black Canary
April Crockett
```

Which hero are you kicking out of your complex?

SUPERHERO NAME: Willy Wonka

Sorry, a hero by the name Willy Wonka could not be found.

moveArrayElements() Function

The `moveArrayElements()` function has the following parameters: a string with the name of the hero that the user wishes to remove, the current number of heroes in the `Heroes` array, and the `Heroes` array. This function returns a `bool` variable that is true if the hero was removed and false if it was not removed.

This function should first find the subscript number of the hero that needs to be removed. Once that is found, then you know there is a hero in the `Heroes` array by that name and that your program will be able to remove it. If your program cannot find the hero in the array, then you return false from this function. Otherwise, this function should now overwrite the element with the hero to delete (x) with the next element in the array (x+1), moving each element after the deleted element to the left. Then return true that the hero was found & removed.

printHeroes() Function

The `printHeroes()` function is a void function and contains the following parameters: current number of heroes in the `Heroes` array and the `Heroes` array. The `printHeroes()` function print each hero to the screen in a nice easy-to-read format.

What would you like to do?

1. Enter some superheroes
2. Delete a superhero
3. Print all superheroes
4. Print rent details.
5. End Program

Enter 1, 2, 3, 4, or 5.

CHOICE: 3

-----SUPERHERO 1

NAME: Superman

DESCRIPTION:

Superman is from Krypton and his weakness is kryptonite. He wears a blue suit with a red S on his chest and a red cape. On Earth, he has super-human abilities such as super-strength.

DANGEROUS? no

RENT PRICE: \$ 1950.85

DAMAGE COST: \$ 200.50

YEARS: 3.00

-----SUPERHERO 2

NAME: Wonder Woman

DESCRIPTION:

Wonder Woman is a founding member of the Justice League, and an ambassador of the Amazon people. She has long dark hair and tall red boots. She was sculpted from clay by her mother Queen Hippolyta and was given a life to live as an Amazon, along with superhuman powers as gifts by the Greek gods. Watch out for her Lasso of Truth and indestructible bracelets!

DANGEROUS? yes

RENT PRICE: \$ 875.00

DAMAGE COST: \$ 300.50

YEARS: 0.50

-----SUPERHERO 3

NAME: Cyborg

DESCRIPTION:

Large portions of Victor Stone's body have been replaced by advanced mechanical parts (hence the name Cyborg), granting him superhuman strength, speed, stamina, and flight. His mechanically-enhanced body, much of which is metallic, is far more durable than a normal human body.

DANGEROUS? no

RENT PRICE: \$ 975.50

DAMAGE COST: \$ 150.45

YEARS: 4.00

-----SUPERHERO 4

NAME: Elongated Man

DESCRIPTION:

As a teenager, Ralph Dibny was fascinated by contortionists, or people who displayed feats of agility and suppleness. He learned that all of the body-benders he spoke with drank a popular soda called "Gingold." Ralph set to work learning chemistry and developed a super-concentrated extract of the rare "gingo" fruit of the Yucatan, which gave him his elasticity.

DANGEROUS? no

RENT PRICE: \$ 350.86

DAMAGE COST: \$ 0.00

YEARS: 10.00

-----SUPERHERO 5

NAME: Zatanna Zatara

DESCRIPTION:

Zatanna makes her living as a stage illusionist prior to discovering her magical abilities while investigating the disappearance of her father. Her original costume is based upon her father's costume but substituting fishnet stockings and high heels for slacks.

DANGEROUS? yes

RENT PRICE: \$ 999.99

DAMAGE COST: \$ 88.88

YEARS: 1.50

-----SUPERHERO 6

NAME: Black Canary

DESCRIPTION:

A world-class martial artist with a superpower: the "canary cry", a high-powered sonic scream which could shatter objects and incapacitate enemies (even ones powerful enough to kill Kryptonians such as Superman).

DANGEROUS? yes

RENT PRICE: \$ 765.15

DAMAGE COST: \$ 44.35

YEARS: 6.00

```

-----SUPERHERO 7
NAME:      April Crockett
DESCRIPTION:
            April is a member of the DC Comics universe first introduced
            in 2002 to help out the Justice League when they were transported
            back in time.  She is telepathic and able to roller skate,
            dance, fight, and sing all at the same time.  She was raised
            to lead a pack of warrior women known as Female Fury Battalion
            but eventually she escapes Apokolips and goes to Earth
            to help Justice League and forms the new Slashing Software
            Skaters group, who skate and code to save Earth.

DANGEROUS?    yes
RENT PRICE:    $ 295.85
DAMAGE COST:   $ 948.33
YEARS:         4.00

```

Notice that the description does word wrapping. In other words, each word is not split up – they are kept together.

You get a bonus of 5 points if you can do word wrapping and indent over your descriptions. Note: since these are bonus points which you earn by doing your own problem solving to figure this out, TAs and instructors are not supposed to help you figure this out!! This is the only part of the program that you can't get help on. It IS possible to implement this feature without including any special libraries or using any functions from libraries with only what you have been taught in CSC 1300.

printRentDetails() function

The **printRentDetails()** function is a void function and it contains two parameters: current number of heroes in the **Heroes** array and the **Heroes** array. This function should print out the monthly rent and damage cost of each superhero in an easy-to-read table format. To get full credit, you MUST use **setw** to create the nice columns.

Also, this function should keep a running total of the total rent cost of all heroes and the total damage cost of all heroes and print this out at the bottom of the table.

```

What would you like to do?
1.  Enter some superheroes
2.  Delete a superhero
3.  Print all superheroes
4.  Print rent details.
5.  End Program
Enter 1, 2, 3, 4, or 5.
CHOICE:  4

```

RENT DETAILS OF EACH HERO:

SUPERHERO	MONTHLY RENT	DAMAGE COST
Superman	\$ 1950.85	\$ 200.50
Wonder Woman	\$ 875.00	\$ 300.50
Cyborg	\$ 975.50	\$ 150.45
Elongated Man	\$ 350.86	\$ 0.00
Zatanna Zatara	\$ 999.99	\$ 88.88
Black Canary	\$ 765.15	\$ 44.35
April Crockett	\$ 295.85	\$ 948.33

TOTALS:	\$ 6213.20	\$ 1733.01

saveToFile() function

The **saveToFile()** function is a void function and it contains two parameters: current number of heroes in the **Heroes** array and the **Heroes** array. The function should ask the user what the name of the file that they wish to save their heroes to. The function should then open that file and write out all the hero's data in the following order:

- Name
- Description
- Dangerous
- Rent cost
- Damage cost
- Number of years at condo

There should be no newlines or endlines in the file. All data should be separated by a pound sign (#) instead of a space.

After printing all data from the Heroes array to the file, this function should print out a message to standard output saying "Your superheroes were successfully saved to the [insert filename here] file."

```
What would you like to do?
1.  Enter some superheroes
2.  Delete a superhero
3.  Print all superheroes
4.  Print rent details.
5.  End Program
Enter 1, 2, 3, 4, or 5.
```

CHOICE: 5

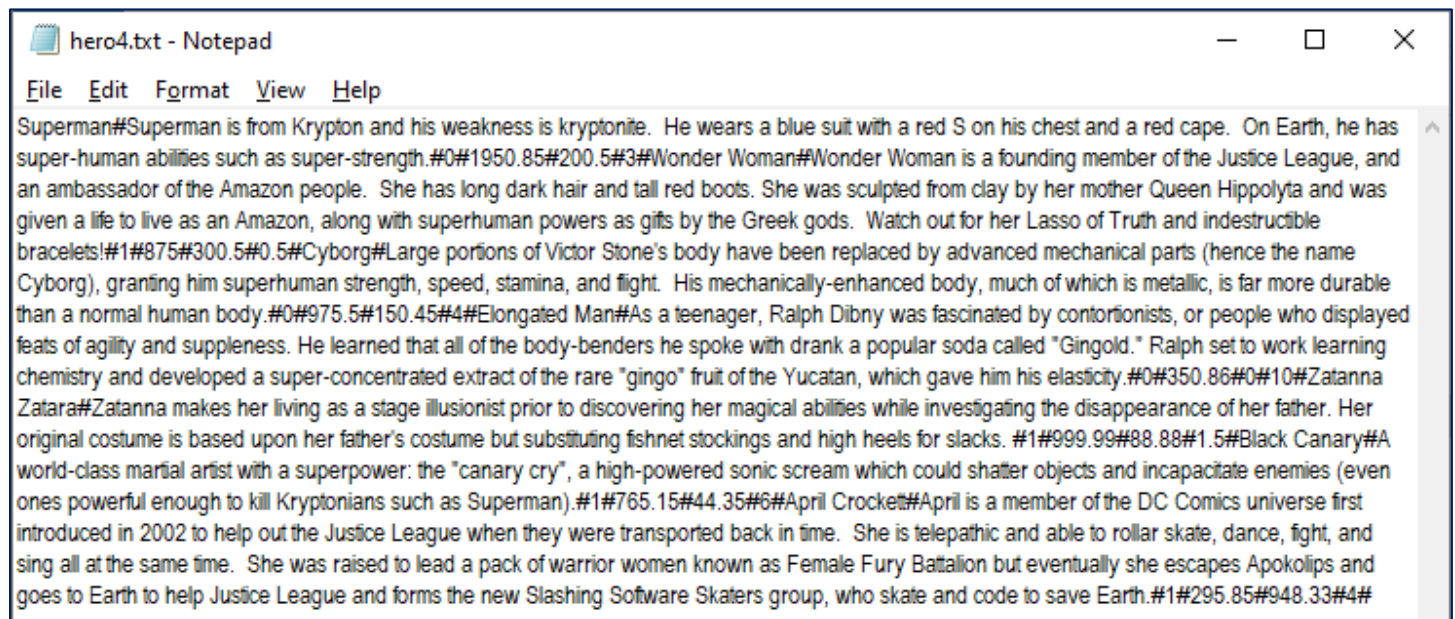
Would you like to save your superheroes list to a file? (y or n) y

What is the name of the file you want to save your creatures to?

FILENAME: hero4.txt

Your superheroes were successfully saved to the hero4.txt file.

GOODBYE!



```
File Edit Format View Help
Superman#Superman is from Krypton and his weakness is kryptonite. He wears a blue suit with a red S on his chest and a red cape. On Earth, he has
super-human abilities such as super-strength.#0#1950.85#200.5#3#Wonder Woman#Wonder Woman is a founding member of the Justice League, and
an ambassador of the Amazon people. She has long dark hair and tall red boots. She was sculpted from clay by her mother Queen Hippolyta and was
given a life to live as an Amazon, along with superhuman powers as gifts by the Greek gods. Watch out for her Lasso of Truth and indestructible
bracelets!#1#875#300.5#0.5#Cyborg#Large portions of Victor Stone's body have been replaced by advanced mechanical parts (hence the name
Cyborg), granting him superhuman strength, speed, stamina, and flight. His mechanically-enhanced body, much of which is metallic, is far more durable
than a normal human body.#0#975.5#150.45#4#Elongated Man#As a teenager, Ralph Dibny was fascinated by contortionists, or people who displayed
feats of agility and suppleness. He learned that all of the body-benders he spoke with drank a popular soda called "Gingold." Ralph set to work learning
chemistry and developed a super-concentrated extract of the rare "gingo" fruit of the Yucatan, which gave him his elasticity.#0#350.86#0#10#Zatanna
Zatara#Zatanna makes her living as a stage illusionist prior to discovering her magical abilities while investigating the disappearance of her father. Her
original costume is based upon her father's costume but substituting fishnet stockings and high heels for slacks. #1#999.99#88.88#1.5#Black Canary#A
world-class martial artist with a superpower: the "canary cry", a high-powered sonic scream which could shatter objects and incapacitate enemies (even
ones powerful enough to kill Kryptonians such as Superman).#1#765.15#44.35#6#April Crockett#April is a member of the DC Comics universe first
introduced in 2002 to help out the Justice League when they were transported back in time. She is telepathic and able to roller skate, dance, fight, and
sing all at the same time. She was raised to lead a pack of warrior women known as Female Fury Battalion but eventually she escapes Apokolips and
goes to Earth to help Justice League and forms the new Slashing Software Skaters group, who skate and code to save Earth.#1#295.85#948.33#4#
```